Experiment 02 — Citation-intent labeling quality

arxivkg · preprocessing · phase 3 (label / bert)

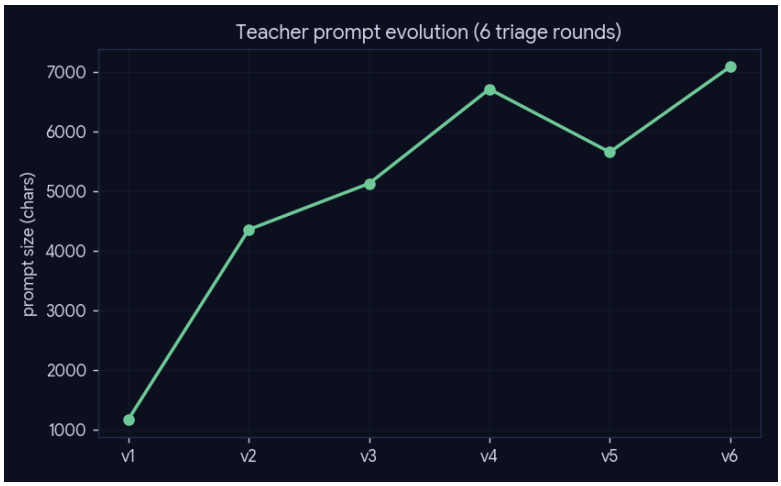
What we did

Phase-3 types every citation edge with an **intent** (6-way: uses / extends / compares / motivates / background / critiques) and a **polarity** (3-way), via a Gemma teacher distilled into a DistilBERT student. We measure labeling quality in two halves: the teacher prompt's six-round evolution, and the distilled student's held-out per-class PRF for both heads.

Method

Pure reads of frozen artifacts (`data/models/student-v*/test_report.json` + the v1–v6 prompts). No model loaded. This is the stage-1 (distilled-label) benchmark; the stage-2 human-gold benchmark comes from the Phase-C labeling tool.

Teacher prompt evolution

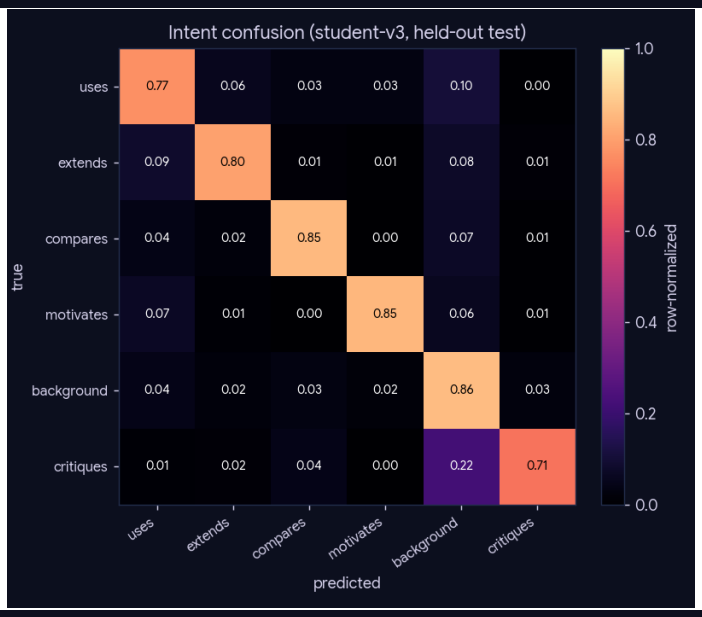


Student quality (held-out test, n = 13,521)

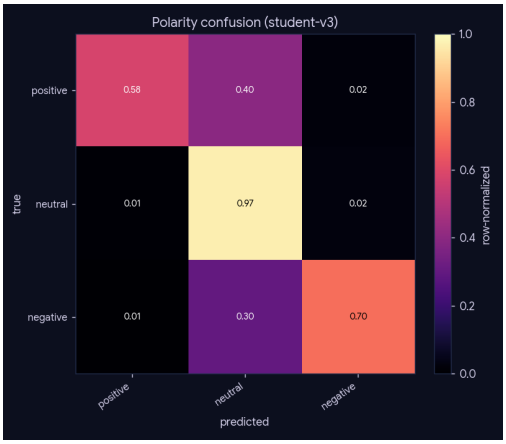
Shipped = **student-v3**: intent acc **0.844** / macro-F1 **0.723**, polarity acc **0.947**.

student	intent acc	intent macro-F1	polarity acc
student-v1	0.894	0.717	0.962
student-v2	0.897	0.713	0.965
student-v3	0.844	0.723	0.947


Intent per-class F1 (rare relational intents lag)
F1
1.0
0.8
0.6
0.4
0.2
0.0
0.69
0.71
0.75
0.70
0.91
0.58
uses
extends
compares
motivates
background
critiques


Intent confusion (student-v3, held-out test)
true
uses 0.77 0.06 0.03 0.03 0.10 0.00
extends 0.09 0.80 0.01 0.01 0.08 0.01
compares 0.04 0.02 0.85 0.00 0.07 0.01
motivates 0.07 0.01 0.00 0.85 0.06 0.01
background 0.04 0.02 0.03 0.02 0.86 0.03
critiques 0.01 0.02 0.04 0.00 0.22 0.71
uses
extends
compares
motivates
background
critiques
predicted
row-normalized
1.0
0.8
0.6
0.4
0.2
0.0


Polarity per-class F1
F1
1.0
0.8
0.6
0.4
0.2
0.0
0.60
0.97
0.58
positive
neutral
negative



Conclusion

Polarity is near-solved (acc 0.95). Intent is strong overall (acc 0.84, macro-F1 0.72) but the rare relational intents (`critiques` 0.58; `motivates`/`extends`/`uses` ∼0.70) lag the dominant `background` (0.91) — exactly the edge types phase-6's tensions/open-problems surfacing depends on. macro-F1 $<$ accuracy reflects this class imbalance. This is a distilled-label benchmark, not human ground truth — those rare classes are what the Phase-C human-gold set must validate.